

Problem A. Lonesome Knight

Time Limit 1000 ms

Mem Limit 65536 kB

OS Windows

The statement of this problem is very simple: you are to determine how many squares of the chessboard can be attacked by a knight standing alone on the board. Recall that a knight moves two squares forward (horizontally or vertically in any direction) and then one square sideways (perpedicularly to the first direction).

Input

The first line contains the number N of test cases, $1 \leq N \leq 64$. Each of the following N lines contains a test: two characters. The first character is a lowercase English letter from 'a' to 'h' and the second character is an integer from 1 to 8; they specify the rank and file of the square at which the knight is standing.

Output

Output N lines. Each line should contain the number of the squares of the chessboard that are under attack by the knight.

Sample

Input	Output
3 a1 d4 g6	2 8 6